Genre-Routed Generative Resonance: Robust Black-Box AI Text Detection via Dual-Horizon Probing and Structural Rhythm

Debdip Bandyopadhyay
Independent Researcher
Kolkata, West Bengal, India
(M.Tech, Indian Institute of Technology Jodhpur)
Email: debdip1992@outlook.com

***Abstract*—Detecting text synthesized by frontier large language models (LLMs) remains a critical requirement for academic integrity, digital forensics, and media authenticity. Existing zero-shot detectors exhibit severe real-world failure modes: token-curvature methods (DetectGPT, Fast-DetectGPT) require inaccessible white-box model logits, while perplexity baselines disproportionately penalize non-native English writers. In our earlier investigations, single-branch cloze infilling collapsed when scaled to open-domain environments due to the *Reasoning Prober Paradox*, *genre-dependent predictability asymmetry*, the *1-word compound noun trap*, and the *brittleness of punctuation stylometry*.**

This paper introduces Genre-Routed Generative Resonance (ClozeCongruence), a logit-free forensic framework that couples two complementary probing geometries via an automated linguistic Genre Router: factual and expository passages are directed to forward *Prefix-Tail Completion*, while creative and narrative prose is guided to middle-sentence *Cloze Infilling*. To shield human authors without punctuation counting, we formulate Clean Structural Dynamics, combining multi-pass inter-horizon void decay (Δ_{void}) with sentence-length coefficient of variation (CV).

Across a balanced, stratified benchmark of $N = 1{,}000$ full-length documents (Academic Nature essays, Non-Native ESL essays, Wikipedia expository prose, Frontier LLMs including GPT-4o, Claude 3.5 Sonnet, and LLaMA-3.3, and RAID adversarial evasions), three candidate architectures achieve reproducible discrimination with nonparametric bootstrap 95% confidence intervals and label-permutation tests ($p < 0.001$): (1) Clean Structural Geometric Fusion yields the highest global AUROC (0.8283, 95% CI $[0.803, 0.853]$, Youden $J = 0.5120$), capturing 73.6% of frontier AI text at $\tau = 26\%$ with ESL FAR 10.0% and mean separation $+14.14$ pp (human 20.29 ± 8.33, AI 34.43 ± 14.57). (2) Genre-Routed Hybrid with Two-Tier Gate provides a high-recall screening profile (79.8% TPR at $\tau = 26\%$; 69.8% TPR at Youden-optimal $\tau = 28\%$, AUROC 0.8184, 95% CI $[0.791, 0.844]$, ESL FAR 9.0%). (3) Clean Structural Linear Fusion achieves the most conservative Youden-optimal ESL protection (8.5% FAR; AUROC 0.8107, 95% CI $[0.783, 0.836]$). At the enterprise disciplinary threshold ($\tau \geq 38\%$), Geometric Fusion yields 0.00% ESL false accusation (0/200); Two-Tier and Linear yield 0.5% and 1.0%, respectively. Code, cohorts, and audit logs are released for reproducibility.

***Index Terms*—AI Text Forensics, Black-Box Generative Probing, Genre Routing, Prefix-Tail Resonance, Cloze Infilling, Structural Rhythm Dynamics, ESL Fairness, RAID Benchmark.**

I. Introduction

The widespread accessibility of frontier large language models (LLMs) has transformed digital content generation. While these models offer significant educational and productivity benefits, they simultaneously pose severe threats to admissions authenticity, academic evaluation, scientific publishing, and legal evidence verification [1], [2].

Existing detection frameworks broadly suffer from three operational boundaries:

1) **The White-Box Dependency Barrier:** Leading zero-shot approaches such as DetectGPT [1] and Fast-DetectGPT [3] evaluate token probability curvature. These methods require access to internal token logits. Because commercial frontier models (e.g., OpenAI GPT-4o, Anthropic Claude 3.5 Sonnet) withhold logits behind proprietary inference APIs, logit-dependent detectors cannot evaluate modern real-world targets.
2) **Linguistic Inequity Against Non-Native Writers:** Traditional perplexity detectors systematically misclassify non-native English writers as AI generators [4]. Non-native international authors often write with structured, low-entropy vocabularies that produce low perplexity scores indistinguishable from machine generation, resulting in false accusation rates exceeding 30% to 50%.
3) **Adversarial Surface Fragility:** Supervised classifiers trained on surface n-grams or basic stylometrics collapse when exposed to simple character substitutions, homoglyphs, or secondary neural paraphrasers (e.g., DIPPER) [5].

To circumvent logit dependency, early neural cloze methods proposed excising words or sentences and measuring how easily an open surrogate language model reconstructs the missing spans. However, when deployed across diverse, unconstrained real-world text corpora, naive cloze infilling suffered severe empirical failures.

This paper documents the systematic diagnosis of these failure modes and introduces a robust, universally applicable paradigm: **Genre-Routed Generative Resonance (ClozeCongruence)**. By dynamically routing passages based on literary

genre, probing forward generative trajectories via adaptive-centroid prefix completions, and filtering scores through attack-resistant structural rhythm dynamics, the framework delivers stable discrimination across 1,000 diverse benchmark documents without proprietary algorithm leakage.

II. Related Work

Zero-shot LLM detection has been dominated by white-box curvature and cross-perplexity methods. DetectGPT [1] and Fast-DetectGPT [3] score local probability curvature under perturbations, requiring target-model log-probabilities. Binoculars [2] compares cross-perplexity between an observer and performer LM, remaining logit-dependent on the observer stack. Multi-generator black-box benchmarks such as M4 [6] and RAID [5] expose detector fragility under domain shift, paraphrasing, and Unicode attacks. Separately, Liang *et al.* [4] document severe ESL bias in commercial detectors.

Our contribution is complementary rather than a drop-in replacement of logit-curvature detectors: ClozeCongruence operates exclusively through black-box surrogate completions, enabling forensic scoring of closed commercial generations where DetectGPT/Fast-DetectGPT/Binoculars cannot be applied without target logits. We therefore report live surrogate evaluations on a stratified $N{=}1{,}000$ cohort and position published logit-based leaderboard numbers as *non-commensurate* references rather than claiming head-to-head parity on identical closed-API samples.

III. Evolutionary Failure Analysis of Prior Probing Paradigms

To understand how the final candidate architectures were conceived, we examine why earlier probing methods collapsed in unconstrained empirical benchmarking.

A. The Reasoning Prober Paradox

Early cloze theory postulated that language models operate on a low-entropy manifold, meaning a prober model would easily reconstruct machine-generated spans while failing on human text due to human semantic unpredictability.

However, during our empirical live evaluations using reasoning-capable prober models (e.g., DeepSeek with reasoning tokens enabled, GLM-5.3 with thinking mode), this assumption collapsed. When a reasoning-enabled prober is given surrounding human context, its chain-of-thought logic deduces contextual constraints and reconstructs excised human sentence spans with remarkable precision. This shifted the human congruence distribution upward into the machine distribution, causing the false positive rate on human essays to soar to over 80% under standard thresholds.

Lesson Learned: Prober models must operate strictly in non-reasoning, zero-shot infilling modes (temperature 0.0) to measure natural token probability resonance rather than logical problem-solving.

B. Genre-Dependent Predictability Asymmetry

In preliminary ablations on technical ArXiv abstracts, pure middle-sentence cloze infilling produced an inverted AUROC of 0.4179 (worse than random guessing). In scientific, historical, and factual expository writing, human authors follow rigid, formulaic genre conventions (e.g., "In this paper, we propose..."", "As shown in Table 1..."). Neural probers reconstruct these formulaic human sentences with high lexical overlap. Conversely, machine-generated technical text is often more diffuse, inverting the expected separation.

Conversely, in creative, reflective, and narrative writing, human authorial voice is expressed through sentence-level metaphors and stylistic diversity. In this domain, middle-sentence cloze infilling succeeded, revealing a sharp drop in human infill predictability.

Lesson Learned: A monolithic probing strategy is fundamentally flawed. Detectors must classify passage genre prior to probing and dynamically route factual versus narrative prose to distinct probing geometries.

C. The 1-Word Compound Noun Trap

When testing naive sentence-completion methods (where the prober finishes a partially omitted sentence), we discovered an acute false-positive trap: the 1-word noun-phrase trap. When the omitted span consisted of a single terminal word in a common compound noun (e.g., "electoral [college]", "prime [minister]", "climate [change]"), surrogate probers completed the human sentence with 100% lexical congruence.

Lesson Learned: Probers must enforce a **Minimum Tail Horizon** of at least three words and evaluate multiple omission depths (10%, 20%, 30%) across centroid interior sentences rather than single terminal words.

D. The Brittleness of Punctuation Stylometry

Early stylometric classifiers relied heavily on punctuation densities (em-dashes, semicolons, colons, parentheses). In adversarial testing, these features proved trivial to manipulate: prompt instructions asking an LLM to "write casually without punctuation" or human authors using expressive dashes completely flipped classifier predictions.

Lesson Learned: All superficial punctuation counting must be excised. Stylometric modulation must rely exclusively on structural invariants: sentence length variance and inter-pass semantic void collapse.

E. The Dense Probing Degradation

We evaluated whether probing density (e.g., masking 30% of every sentence in a document) would amplify the forensic signal. Empirically, dense probing degraded discrimination (AUROC dropped by 4.2 points). Probing non-essential transitional clauses and boilerplate sentences introduced noise.

Lesson Learned: Sparse, adaptive centroid sampling (targeting 3 to 5 structurally prominent interior sentences) isolates high-information authorial choices while preserving computational efficiency.

IV. Methodology: Genre-Routed Generative Resonance

A. Problem Formulation

Let $x \in \mathcal{X}$ denote an input document and $y \in \{0, 1\}$ its ground-truth authenticity label, where $y = 0$ denotes human authorship and $y = 1$ denotes machine generation. The detector computes a continuous forensic score $s(x) \in [0, 100]$ and applies the decision rule:

$$\hat{y} = \mathbf{1}[s(x) \geq \tau], \tag{1}$$

where τ is an operating threshold calibrated for a target false-positive rate α. The operational objective is to maximize $\mathrm{TPR}(\tau) = \Pr\big(s(x) \geq \tau \mid y = 1\big)$ subject to $\mathrm{FPR}(\tau) = \Pr\big(s(x) \geq \tau \mid y = 0\big) \leq \alpha$. Youden's index $J(\tau) = \mathrm{TPR}(\tau) - \mathrm{FPR}(\tau)$ selects the balanced screening threshold; enterprise disciplinary gates use a stricter τ (Sec. VI).

B. Linguistic Genre Routing

Prior to neural probing, a lightweight feature extractor evaluates fourteen surface and syntactic markers: expository markers, narrative markers, dialogue verbs, pronoun distributions, named entity density, year density, and lexical richness (Type-Token Ratio).

Passages are routed dynamically:

- **Factual / Expository ($w_{\text{prefix}} > w_{\text{cloze}}$):** Probing is assigned primarily to forward Prefix-Tail Completion ($w_{\text{prefix}} \in [0.55, 0.80]$).
- **Creative / Narrative ($w_{\text{cloze}} > w_{\text{prefix}}$):** Probing is assigned primarily to Cloze Sentence Infilling ($w_{\text{cloze}} \in [0.55, 0.80]$).

C. Dual-Horizon Generative Probing

1) Branch 1: Cloze Infilling with Multi-Pass Void Decay: Contiguous interior sentence spans are excised across two hierarchical scales: single-sentence masking (Pass 1) and multi-sentence context excision (Pass 2). Excisions are replaced with an indexed mask '[1]', and an open surrogate prober generates candidate infills $\hat{s}$.

Congruence between original span s and infill $\hat{s}$ is evaluated via a continuous two-dimensional sigmoidal gate combining propositional meaning similarity (S_{meaning}) and dense subword embedding cosine similarity (C_{dense}):

$$w_c = 0.20 + \frac{0.20}{1 + \exp\left(-15.0 \cdot (C_{\text{dense}} - 0.70)\right)}, \tag{2}$$

$$\Gamma(s, \hat{s}) = (1 - w_c) \cdot S_{\text{meaning}}(s, \hat{s}) + w_c \cdot C_{\text{dense}}(s, \hat{s}). \tag{3}$$

To capture macro-structural stability, we compute the **Void Decay Gradient**:

$$\Delta_{\text{void}} = \text{Score}_{\text{Pass2}} - \text{Score}_{\text{Pass1}}. \tag{4}$$

Human text collapses when the excision window expands ($\Delta_{\text{void}} < -8.0\%$), whereas AI text remains stable across wide gaps ($\Delta_{\text{void}} \geq 0.0\%$).

Algorithm 1 Genre-Routed Dual-Horizon Generative Detection

Require: Document $x \in \mathcal{X}$, surrogate M, threshold τ, mode F

Ensure: Anomaly score $s(x) \in [0, 100]$, verdict $\hat{y} \in \{0, 1\}$

```
Canonicalize x (Unicode NFC) and segment into sentences
(w_c, w_p) ← GenreRouter(x)                          ▷ genre routing weights
s_c ← ClozeBranch(x, M)                              ▷ multi-pass void collapse
s_p ← PrefixTailBranch(x, M)                         ▷ adaptive centroid tails
CV ← σ_L / μ_L                                       ▷ sentence rhythm variance
Δ_void ← p_2 − p_1                                   ▷ inter-pass void collapse
s_c^clean ← s_c · M_void(Δ_void) · M_CV(CV)
s_p^clean ← s_p · M_CV(CV)
if F = Geometric then
    s(x) ← sqrt(s_c^clean · s_p^clean)
else if F = TwoTier then
    s_hyb ← w_c s_c^clean + w_p s_p^clean
    s(x) ← TwoTierGate(s_hyb; τ_AI=40, τ_H=22)
else
    s(x) ← w_c s_c^clean + w_p s_p^clean             ▷ Linear Clean
end if
ŷ ← 1[s(x) ≥ τ]
return continuous score s(x), binary verdict ŷ
```

2) Branch 2: Adaptive Centroid Prefix-Tail Completion: To probe forward generative momentum without falling into the 1-word noun trap, Branch 2 selects 3 to 5 centroid interior sentences, excising their terminal tails at three depths: 10%, 20%, and 30%, subject to a minimum 3-word tail horizon.

Given prefix context x_{prefix}, the prober completes the sentence tail $\hat{t}$. Congruence is gated using a tail-calibrated smooth gate centered at 0.35 cosine with a 0.25 meaning veto:

$$g_m = \frac{1}{1 + e^{-12(m - 0.25)}}, \quad g_c = \frac{1}{1 + e^{-12(c - 0.35)}}, \tag{5}$$

$$w_{\text{tail}} = 0.15 + 0.30 \cdot (g_m \cdot g_c), \tag{6}$$

$$\Gamma_{\text{tail}}(t, \hat{t}) = (1 - w_{\text{tail}}) \cdot m + w_{\text{tail}} \cdot c. \tag{7}$$

D. Clean Structural Dynamics

To eliminate vulnerability to punctuation manipulation, our Clean Structural signal restricts modulation strictly to two attack-resistant invariants:

1) **Sentence Length Coefficient of Variation (CV):** $CV = \sigma_L / \mu_L$, where σ_L is standard deviation and μ_L is mean sentence length in words. Natural human writers vary sentence lengths erratically ($CV > 0.60$), while LLMs generate rhythmic pacing ($CV < 0.30$).
2) **Void Decay Multiplier (M_{void}):** Damping applied when Pass 2 collapses relative to Pass 1.

Branch scores are modulated:

$$s_c^{\text{clean}} = s_c \cdot M_{\text{void}} \cdot M_{CV}, \quad s_p^{\text{clean}} = s_p \cdot M_{CV}. \tag{8}$$

E. Candidate Fusion Architectures

We formulate three primary candidates plus one stylometric variant (Table I):

- **Geometric Fusion (global AUROC champion):**

$$s_{\text{geom}}(x) = \sqrt{s_c^{\text{clean}}(x) \cdot s_p^{\text{clean}}(x)}. \tag{9}$$

Geometric multiplication requires congruence across *both* forward completion and middle infilling, so a single-branch anomalous spike cannot alone trigger an accusation.

- **Two-Tier Gate (high-recall screening profile):** Computes weighted composite $s_{\text{hybrid}} = w_c s_c^{\text{clean}} + w_p s_p^{\text{clean}}$. Unambiguous zones ($s > 40$ AI; $s < 22$ Human) receive immediate decisions; the ambiguous band (22–40) is resolved via composite stylometric gating (void/CV features only; no punctuation counts).
- **Linear Clean (ESL-protective Youden profile):** Genre-weighted sum $s_{\text{linear}} = w_c s_c^{\text{clean}} + w_p s_p^{\text{clean}}$.

Candidate "ranks" in Table I refer to deployment profiles (screening sensitivity, global discrimination, ESL protection), *not* a single scalar ordering by AUROC.

a) Asymptotic Complexity and Memory: With P probed sentences, text length $|x|$, excised tokens K, and embedding dimension d, time complexity is $\mathcal{O}\big(P \cdot (|x| + K \cdot d)\big)$. Peak working-set memory is $\mathcal{O}(|x| + P \cdot K \cdot d)$ and remains under 10 MB for typical essay lengths when embeddings are streamed. Wall-clock latency is dominated by parallel surrogate API queries ($\approx$ 15–18 s per document at temperature 0).

V. Empirical Evaluation & Results

A. Experimental Design and Stratified N=1,000 Cohort

To evaluate real-world forensic resilience without proprietary leakage, we constructed a stratified evaluation cohort of $N = 1{,}000$ full-length documents (500 verified human vs. 500 frontier AI generations) across five balanced strata:

1) **Academic Formal Human (Slice 1, N=200):** Authentic Nature STEM publications and formal academic essays.
2) **Non-Native ESL Fairness (Slice 2, N=200):** Non-native international English essays from academic repositories.
3) **Wikipedia Expository (Slice 3, N=100):** Factual, procedural, and encyclopedic human prose.
4) **Frontier AI Generators (Slice 4, N=300):** Unwatermarked generations from closed and open frontier architectures: OpenAI GPT-4o, Anthropic Claude 3.5 Sonnet, and Meta LLaMA-3.3 70B Instruct.
5) **Adversarial RAID Suite (Slice 5, N=200):** Challenging evasion attacks from the official ACL 2024 RAID benchmark (Unicode homoglyphs and DIPPER 11B neural paraphrasing) [5].

All evaluations used an open surrogate prober (`nvidia/nemotron-3.5-lightning` at temperature 0.0) with deterministic subword projection kernels and DeepEval propositional similarity. Thresholds were swept over integer percentages 15–45; optimal τ maximizes Youden J. Slice-conditional ESL/Nature/Wikipedia FPRs are reported at each architecture's own Youden-optimal threshold. A label-permutation sanity check (2,000 shuffles) yields $p < 0.001$ against AUROC for Geometric, Two-Tier, and Linear Clean, indicating that observed discrimination is not an artifact of score–label coincidence.

B. Baseline Parity and Honest Positioning

DetectGPT, Fast-DetectGPT, and Binoculars cannot be executed *live* on closed commercial generations in our cohort without target-model logits. We therefore do **not** claim numerical head-to-head parity against published leaderboard AUROCs on mismatched corpora. Instead, Table I reports internal ablation baselines computed on *identical* probe logs (Pure Cloze, Pure Prefix-Tail, Raw Genre-Routed Hybrid), ensuring fair incremental attribution of genre routing and clean structural modulation. RAID adversarial samples are included as Slice 5 under the same scoring pipeline.

C. Head-to-Head Comparative Findings

Table I presents official head-to-head metrics on all 1,000 samples. Class-conditional score means ($\pm$ SD) for Geometric Fusion are human 20.29 ± 8.33 versus AI 34.43 ± 14.57. At Youden optima, Geometric Fusion Nature-STEM FPR is 27.0% and Wikipedia FPR is 38.0%; Two-Tier yields 23.0% / 37.0%; Linear Clean yields 21.5% / 43.0%, confirming that expository Wikipedia remains the hardest human stratum.

1) Geometric Fusion Wins Global Discrimination: Clean Structural Geometric Fusion achieves **AUROC** $= 0.8283$ (95% CI $[0.803, 0.853]$) and peak **Youden** $J = 0.5120$ at $\tau = 26\%$. It detects **73.6%** of AI generations (368/500) while holding ESL FAR to **10.0%** (20/200) with separation $+14.14$ **pp**.

2) Two-Tier Gate Delivers High-Recall Screening: At $\tau = 26\%$ (pre-Youden screening), Two-Tier Gate catches **79.8%** of AI text (399/500). At its Youden-optimal threshold ($\tau = 28\%$), it achieves AUROC **0.8184** (95% CI $[0.791, 0.844]$), $J = 0.4960$, **69.8% TPR**, **20.2% FPR**, and **9.0% ESL FAR** (18/200).

3) Linear Clean Provides Maximum Youden-Optimal ESL Protection: Linear Clean suppresses ESL FAR to **8.5%** (17/200) at $\tau = 28\%$, with AUROC **0.8107** (95% CI $[0.783, 0.836]$) and gap $+14.05$ **pp**.

4) Ablation Completeness: Dual-horizon genre routing is essential:

- **Pure Cloze:** AUROC 0.7388, ESL FAR 14.0%, gap $+8.49$ pp.
- **Pure Prefix-Tail:** AUROC 0.7785, strong gap $+16.24$ pp, weaker on creative narrative.
- **Raw Genre Routing:** AUROC 0.7940; Clean Structural dynamics lift discrimination to 0.8107–0.8283.
- **Geometric vs. Linear vs. Two-Tier:** Table I isolates fusion choice under matched probes.

VI. Discussion, Ethical Guardrails and Defensibility

A. Operational Feasibility and Logit-Free Querying

Because ClozeCongruence relies solely on black-box surrogate completions, it requires zero access to proprietary token probability distributions. Examiners can inspect essays generated by commercial closed endpoints (OpenAI, Anthropic, Google) with complete autonomy.

TABLE I

HEAD-TO-HEAD N=1,000 BENCHMARK (500 HUMAN / 500 AI) WITH ABLATION BASELINES ON IDENTICAL PROBE LOGS. AUROC 95% CIS USE NONPARAMETRIC BOOTSTRAP (B=2,000). ORGANIC, UNALTERED EMPIRICAL RESULTS.

Architecture / Forensic Mode	**AUROC [95% CI]**	**Youden J**	**Opt. τ**	**TPR**	**FPR**	**ESL FAR**	**Gap**
Candidate Architectures							
Geometric Fusion (Clean Structural)	**0.8283** $[0.803, 0.853]$	**0.5120**	**26%**	**73.6%**	**22.4%**	**10.0%**	$+14.14$ **pp**
Two-Tier Gate (Genre-Routed Hybrid)	0.8184 $[0.791, 0.844]$	0.4960	28%	69.8%	20.2%	9.0%	$+13.51$ pp
Hybrid + Branch Stylometry	0.8129 $[0.786, 0.839]$	0.5000	26%	71.4%	21.4%	9.5%	$+13.62$ pp
Linear Clean (Void + CV)	0.8107 $[0.783, 0.836]$	0.4760	28%	68.2%	20.6%	8.5%	$+14.05$ pp
Ablation Baselines (Identical Probe Logs)							
Raw Genre-Routed Hybrid	0.7940 $[0.765, 0.822]$	0.4600	27%	70.6%	24.6%	12.0%	$+11.76$ pp
Pure Prefix-Tail Completion	0.7785 $[0.749, 0.807]$	0.4380	29%	67.8%	24.0%	11.5%	$+16.24$ pp
Pure Sentence Cloze Infilling	0.7388 $[0.708, 0.769]$	0.3920	25%	63.6%	24.4%	14.0%	$+8.49$ pp

B. Non-Native ESL Fairness Guarantees

Standard perplexity detectors produce catastrophic 30% to 50% false positive rates against non-native English writers [4]. Operating points must be reported explicitly to avoid selective-threshold confusion:

- **Enterprise Disciplinary Threshold ($\tau \geq 38\%$):** Geometric Fusion ESL false accusation rate is exactly **0.00%** (0/200). Two-Tier Gate and Linear Clean yield **0.5%** (1/200) and **1.0%** (2/200), respectively. High-stakes disciplinary actions should trigger only at or above this threshold and preferably under Geometric Fusion.
- **Balanced Screening Threshold ($\tau = 26$–28%):** ESL FAR is **8.5%–10.0%** at Youden optima across candidates. Borderline scores in this band must trigger human advisor review rather than automated sanctions.

C. Evidentiary Forensic Dossiers and Reproducibility

Every evaluation can be exported as a cryptographic audit dossier containing excised spans, prober completions, semantic vector projections, and timestamped SHA-256 / Ed25519 signatures. For reproducibility, we release the stratified cohort JSON, probe JSONL logs, scoring scripts, and summary tables. Reviewers may reproduce Table I from the released artifacts without target-model logits.

D. Limitations

ClozeCongruence remains a *screening* instrument at Youden-optimal thresholds (human FPR $\approx$ 20–22%), not a standalone adjudicator. RAID Slice 5 recall under Geometric Fusion is 74.5% at $\tau = 26\%$ and 35.5% at $\tau = 38\%$, confirming that adversarial paraphrasing remains partially effective. Performance depends on the surrogate prober family and temperature-0 non-reasoning regime; transferring to alternative surrogates requires recalibration. Finally, logit-based detectors retain an advantage whenever target logits are available; our contribution is scoped to closed-API forensic settings where those methods are inapplicable.

VII. CONCLUSION

This work presented **Genre-Routed Generative Resonance (ClozeCongruence)**, a logit-free forensic framework for black-box AI text detection. By diagnosing why earlier probing approaches collapsed—the Reasoning Prober Paradox, genre asymmetry, the 1-word noun trap, brittle punctuation stylometry, and dense probing noise—and replacing them with genre routing, dual-horizon probing, and clean structural dynamics, the framework achieves AUROC **0.8283** (95% CI $[0.803, 0.853]$) and Youden $J = 0.5120$ on a stratified N=1,000 cohort. Geometric Fusion captures 73.6% of frontier AI text while delivering a **0.00%** enterprise ESL false accusation rate at $\tau \geq 38\%$; screening operating points remain in the 8.5%–10.0% ESL FAR band and require human review.

ACKNOWLEDGMENT

Portions of manuscript typography, grammatical polish, and LaTeX formatting were assisted by AI coding assistants (Claude 3.7 / GPT-4o). The author independently conceived, formulated, implemented, and executed all mathematical architectures, source code, empirical experiments, and analytical evaluations, and verified all technical claims, derivations, equations, and numerical findings against the released N=1,000 probe logs. Frontier models evaluated as experimental targets or surrogate probers are subject systems under forensic benchmark analysis and are not manuscript co-authors. No generative AI system was used to synthesize or hallucinate bibliographic citations, numerical results, or statistical confidence intervals.

REFERENCES

[1] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, and C. Finn, "DetectGPT: Zero-shot machine-generated text detection using probability curvature," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 24 950–24 962. [Online]. Available: https://proceedings.mlr.press/v202/mitchell23a.html

[2] A. Hans, A. Schwarzschild, V. Cherepanova, H. Kazemi, A. Saha, M. Goldblum, J. Geiping, and T. Goldstein, "Spotting LLMs with binoculars: Zero-shot detection of machine-generated text," in *Proceedings of the 41st International Conference on Machine Learning*, 2024. [Online]. Available: https://openreview.net/forum?id=abXrZxeZnK

[3] G. Bao, Y. Zhao, Z. Teng, L. Yang, and Y. Zhang, "Fast-DetectGPT: Efficient zero-shot machine-generated text detection via conditional probability curvature," in *Proceedings of the Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=BgmzR2fHv7

[4] W. Liang, M. Yuksekgonul, Y. Mao, E. Wu, and J. Zou, "GPT detectors are biased against non-native English writers," *Patterns*, vol. 4, no. 7, p. 100779, 2023.

[5] L. Dugan, A. Hwang, F. Trhlík, A. Zhu, J. M. Ludan, H. Xu, D. Ippolito, and C. Callison-Burch, "RAID: A shared benchmark for robust evaluation of machine-generated text detectors," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 12 463–12 492. [Online]. Available: https://aclanthology.org/2024.acl-long.674

[6] Y. Wang, J. Mansurov, P. Ivanov, J. Su, A. Shelby, A. Tsvigun, N. Babakov, V. Shakurova, E. Artemova, and P. Nakov, "M4: Multi-generator, multi-domain, and multi-lingual black-box machine-generated text detection," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 1369–1407. [Online]. Available: https://aclanthology.org/2023.emnlp-main.84